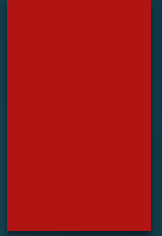


Customer Churn Prediction for SyriaTel



Business Problem

- ▶ Telecommunication companies face revenue loss when customers leave (churn).
- ▶ SyriaTel wants to predict which customers are likely to churn.
- ▶ Goals:
 - ▶ - Predict churn customers
 - ▶ - Identify key factors influencing churn
 - ▶ - Enable proactive retention strategies

Dataset Overview

- ▶ Dataset: Telecom Customer Churn Dataset
- ▶ Total customers: 3,333
- ▶ Target variable: Churn (Yes/No)

- ▶ Key features include:
 - ▶ - Call minutes and call counts
 - ▶ - International plan
 - ▶ - Customer service calls
 - ▶ - Voice mail plan

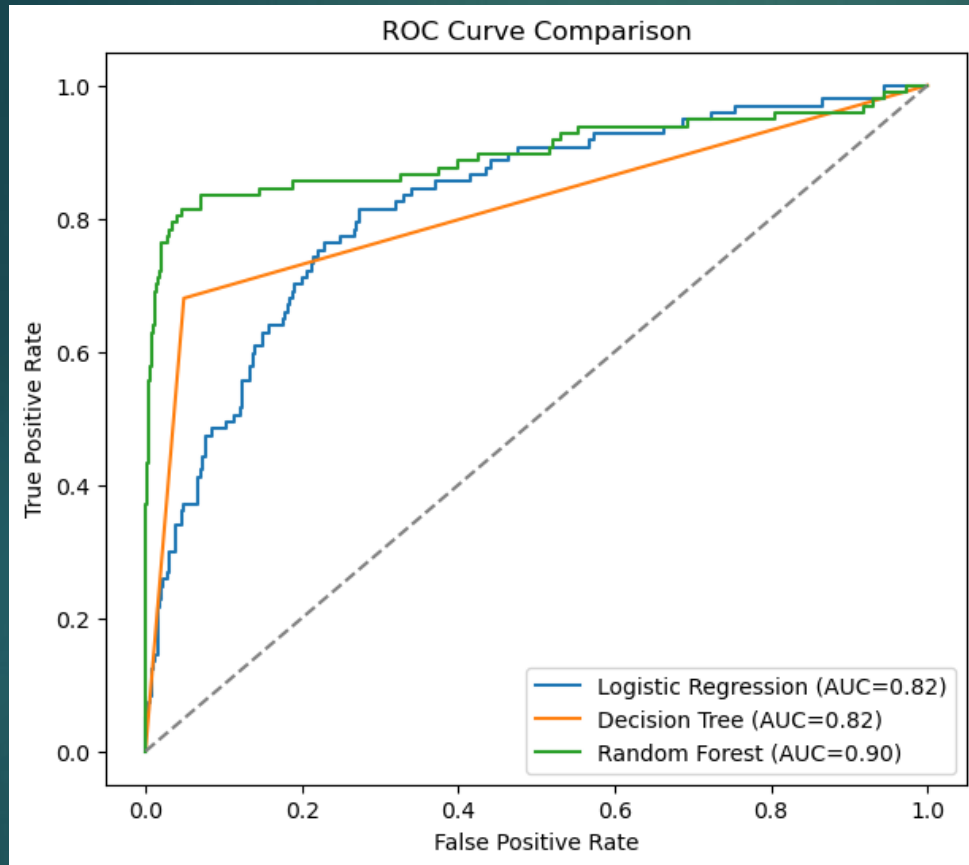
Modeling Approach

- ▶ Steps performed:
 - ▶ - Data cleaning and preprocessing
 - ▶ - Feature engineering
 - ▶ - Train-test split
 - ▶ - Pipeline implementation
- ▶ Models evaluated:
 - ▶ - Logistic Regression
 - ▶ - Decision Tree
 - ▶ - Random Forest

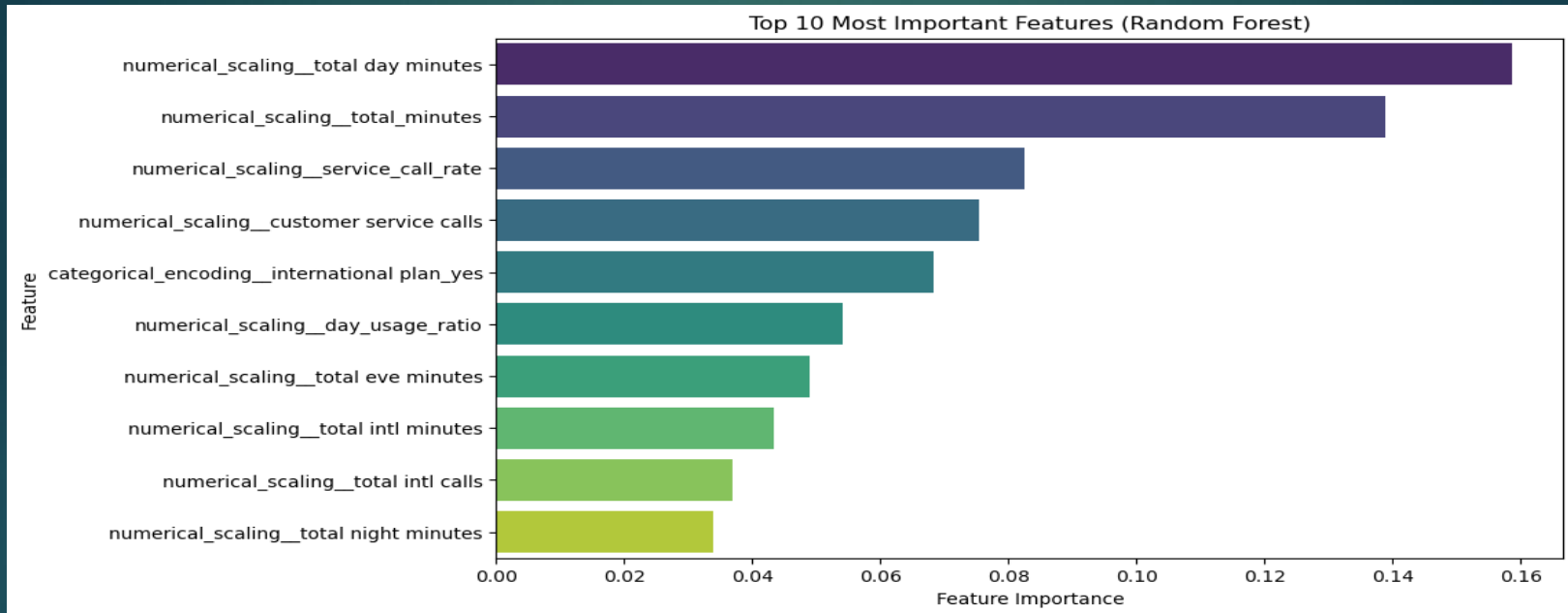
Model Performance Comparison

- ▶ Logistic Regression
 - ▶ - Accuracy: 0.87
 - ▶ - Churn Recall: 0.30
 - ▶ - AUC: 0.82
- ▶ Decision Tree
 - ▶ - Accuracy: 0.91
 - ▶ - Churn Recall: 0.68
 - ▶ - AUC: 0.82
- ▶ Random Forest
 - ▶ - Accuracy: 0.92
 - ▶ - Churn Recall: 0.45
 - ▶ - AUC: 0.90

ROC Curve Comparison



Features Visualizations



Key Insights

- ▶ - Random Forest achieved the highest overall accuracy (92%) and AUC (90%).
- ▶ - Decision Tree detected the most churn customers (highest recall).
- ▶ - Logistic Regression provided a strong interpretable baseline.
- ▶ Customer service calls and high usage patterns were strong churn indicators.

Business Recommendations

- ▶ SyriaTel should implement churn prediction systems to identify high-risk customers.
- ▶ Recommended actions:
 - ▶ - Target high-risk customers with retention offers
 - ▶ - Improve customer service response
 - ▶ - Monitor heavy daytime usage customers
 - ▶ - Continuously retrain models with new customer data

Conclusion

- ▶ Machine learning can effectively predict customer churn.
- ▶ Random Forest provided the strongest predictive performance.
- ▶ Using predictive analytics, SyriaTel can reduce churn,
- ▶ Improve customer satisfaction, and increase long-term revenue.